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Phase and amplitude error correction in a transmission circuit

Abstract

Phase and amplitude error correction in a transmission circuit is provided. The transmission circuit includes a transceiver circuit, a power management integrated circuit (PMIC), and a power amplifier circuit(s). The transceiver circuit generates a radio frequency (RF) signal(s) from an input vector, the PMIC generates a modulated voltage, and the power amplifier circuit(s) amplifies the RF signal(s) based on the modulated voltage. When the power amplifier circuit(s) is coupled to an RF front-end circuit, unwanted amplitude-amplitude (AM-AM) and amplitude-phase (AM-PM) errors may be created across a modulation bandwidth of the transmission circuit. In this regard, in embodiments disclosed herein, the input vector is equalized based on multiple complex filters to thereby cause the AM-AM and AM-PM errors to be corrected in the transmission circuit. As a result, it is possible to reduce undesired instantaneous excessive compression and/or spectrum regrowth across the modulation bandwidth of the transmission circuit.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of provisional patent application Ser. No. 63/245,160, filed Sep. 16, 2021, and provisional patent application Ser. No. 63/245,151, filed Sep. 16, 2021, the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE

(1) The technology of the disclosure relates generally to correcting amplitude-amplitude (AM-AM) and amplitude-phase (AM-PM) errors in a transmission circuit.

BACKGROUND

(2) Mobile communication devices have become increasingly common in current society for providing wireless communication services. The prevalence of these mobile communication devices is driven in part by the many functions that are now enabled on such devices. Increased processing capability in such devices means that mobile communication devices have evolved from being pure communication tools into sophisticated mobile multimedia centers that enable enhanced user experiences.

(3) The redefined user experience relies on a higher data rate offered by advanced fifth generation (5G) and 5G new radio (5G-NR) technologies, which typically transmit and receive radio frequency (RF) signals in millimeter wave spectrums. Given that the RF signals are more susceptible to attenuation and interference in the millimeter wave spectrums, the RF signals are typically amplified by state-of-the-art power amplifiers to help boost the RF signals to higher power before transmission.

(4) In a typical transmission circuit, a transceiver circuit is configured to generate an RF signal, a power management circuit is configured to generate a modulated voltage, a power amplifier circuit is configured to amplify the RF signal based on the modulated voltage, and an antenna circuit is configured to transmit the RF signal in one or more transmission frequencies. The power amplifier circuit can be further coupled to the antenna circuit via an RF front-end circuit (e.g., filter, switches, etc.). Notably, an output reflection coefficient (e.g., $S_{sub.22}$) of the power amplifier circuit can interact with an input reflection coefficient (e.g., $S_{sub.11}$) of the RF front-end circuit to cause a group delay in the RF signal to potentially create an amplitude-to-amplitude (AM-AM) error in the modulated voltage and/or an amplitude-to-phase (AM-PM) error in the RF signal. As such, it is desirable to correct the AM-AM and AM-PM errors in all of the transmission frequencies to help prevent undesired amplitude distortion and/or spectrum regrowth, particularly when the RF signal is modulated across a wide modulation bandwidth (e.g., ≥ 200 MHz).

SUMMARY

(5) Embodiments of the disclosure relate to phase and amplitude error correction in a transmission circuit. The transmission circuit includes a transceiver circuit, a power management integrated circuit (PMIC), and a power amplifier circuit(s). The transceiver circuit generates a radio frequency (RF) signal(s) from an input vector, the PMIC generates a modulated voltage, and the power amplifier circuit(s) amplifies the RF signal(s) based on the modulated voltage. When the power amplifier circuit(s) is coupled to an RF front-end circuit (e.g., filter/multiplexer), an output reflection coefficient (e.g., $S_{sub.22}$) of the power amplifier circuit(s) can interact with an input reflection coefficient (e.g., $S_{sub.11}$) of the RF front-end circuit to create unwanted amplitude-amplitude (AM-AM) and amplitude-phase (AM-PM) errors across a modulation bandwidth of the transmission circuit. In this regard, in embodiments disclosed herein, the input vector is equalized based on multiple complex filters to thereby cause the AM-AM and AM-PM errors to be corrected in the transmission circuit. As a result, it is possible to reduce undesired instantaneous excessive compression and/or spectrum regrowth across the modulation bandwidth of the transmission circuit.

(6) In one aspect, a transmission circuit is provided. The transmission circuit includes a PMIC. The PMIC includes a phase correction circuit. The phase correction circuit is configured to receive a modulated target voltage and an indication of a selected target frequency among multiple target frequencies. The phase correction circuit is also configured to determine a reference phase correction voltage corresponding to a reference frequency based on the modulated target voltage. The phase correction circuit is also configured to determine a scaling factor corresponding to the selected target frequency. The phase correction circuit is also configured to generate the phase correction voltage based on the determined reference phase correction voltage and the determined scaling factor. The PMIC also includes a power amplifier circuit. The power amplifier circuit includes a phase shifter. The phase shifter is configured to determine a phase shift based on the phase correction voltage. The phase shifter is also configured to apply the phase shift to an RF

signal modulated for transmission in the selected target frequency to thereby generate a phase-shifted RF signal.

(7) In another aspect, a method for correcting phase and amplitude errors in a transmission circuit is provided. The method includes receiving a modulated target voltage and an indication of a selected target frequency among multiple target frequencies. The method also includes determining a reference phase correction voltage corresponding to a reference frequency based on the modulated target voltage. The method also includes determining a scaling factor corresponding to the selected target frequency. The method also includes generating a phase correction voltage based on the determined reference phase correction voltage and the determined scaling factor. The method also includes determining a phase shift based on the phase correction voltage. The method also includes applying the phase shift to an RF signal modulated for transmission in the selected target frequency to thereby generate a phase-shifted RF signal.

(8) In another aspect, a PMIC is provided. The PMIC includes a phase correction circuit. The phase correction circuit is configured to receive a modulated target voltage and an indication of a selected target frequency among multiple target frequencies. The phase correction circuit is also configured to determine a reference phase correction voltage corresponding to a reference frequency based on the modulated target voltage. The phase correction circuit is also configured to determine a scaling factor corresponding to the selected target frequency. The phase correction circuit is also configured to generate a phase correction voltage based on the determined reference phase correction voltage and the determined scaling factor. The PMIC also includes a voltage modulation circuit. The voltage modulation circuit is configured to generate a modulated voltage based on the modulated target voltage.

(9) Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.

(2) FIG. 1A is a schematic diagram of an exemplary existing transmission circuit, wherein unwanted amplitude-amplitude (AM-AM) and amplitude-phase (AM-PM) errors may be created when a power amplifier circuit is coupled to a radio frequency (RF) front-end circuit;

(3) FIG. 1B is a schematic diagram providing an exemplary illustration of an output stage of the power amplifier circuit in FIG. 1A;

(4) FIG. 1C is a schematic diagram of an exemplary equivalent model providing an exemplary illustration of the unwanted voltage distortion filter created by a coupling between the power amplifier circuit and the RF front-end circuit 14 in FIG. 1A;

(5) FIG. 1D is a graphic diagram illustrating a distribution of multiple group delays across multiple transmission frequencies;

(6) FIG. 1E is a graphic diagram illustrating a distribution of multiple variable phase errors across multiple transmission frequencies;

(7) FIG. 1F is a graphic diagram illustrating a distribution of the AM-AM and AM-PM errors across multiple transmission frequencies;

(8) FIG. 2 is a schematic diagram of an exemplary transmission circuit that can be configured according to embodiments of the present disclosure to correct the AM-AM and the AM-PM errors across a modulation bandwidth that includes multiple transmission frequencies;

- (9) FIG. 3 is a schematic diagram illustrating a transceiver circuit in the transmission circuit of FIG. 2 configured according to an embodiment of the present disclosure;
- (10) FIG. 4A is a graphic diagram illustrating a distribution of multiple constant group delays across multiple transmission frequencies;
- (11) FIG. 4B is a graphic diagram illustrating a distribution of multiple linearly related variable phase errors across the multiple transmission frequencies in FIG. 4A;
- (12) FIG. 5 is a schematic diagram illustrating a scaling circuit in the transmission circuit of FIG. 2 configured according to an embodiment of the present disclosure; and
- (13) FIG. 6 is a flowchart of an exemplary process that can be performed by the transmission circuit of FIG. 2 to correct the AM-AM and AM-PM errors in the transmission circuit.

DETAILED DESCRIPTION

- (14) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.
- (15) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
- (16) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.
- (17) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.
- (18) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
- (19) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this

disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(20) Embodiments of the disclosure relate to phase and amplitude error correction in a transmission circuit. The transmission circuit includes a transceiver circuit, a power management integrated circuit (PMIC), and a power amplifier circuit(s). The transceiver circuit generates a radio frequency (RF) signal(s) from an input vector, the PMIC generates a modulated voltage, and the power amplifier circuit(s) amplifies the RF signal(s) based on the modulated voltage. When the power amplifier circuit(s) is coupled to an RF front-end circuit (e.g., filter/multiplexer), an output reflection coefficient (e.g., $S_{\text{sub.22}}$) of the power amplifier circuit(s) can interact with an input reflection coefficient (e.g., $S_{\text{sub.11}}$) of the

(21) RF front-end circuit to create unwanted amplitude-amplitude (AM-AM) and amplitude-phase (AM-PM) errors across a modulation bandwidth of the transmission circuit. In this regard, in embodiments disclosed herein, the input vector is equalized based on multiple complex filters to thereby cause the AM-AM and AM-PM errors to be corrected in the transmission circuit. As a result, it is possible to reduce undesired instantaneous excessive compression and/or spectrum regrowth across the modulation bandwidth of the transmission circuit.

(22) Before discussing the transmission circuit according to the present disclosure, starting at FIG. 2, a brief discussion of an existing transmission circuit is first provided to help understand how unwanted AM-AM and AM-PM errors may be created when a power amplifier circuit is coupled to an RF front-end circuit, such as a filter/multiplexer circuit.

(23) FIG. 1A is a schematic diagram of an exemplary existing transmission circuit **10**, wherein an unwanted voltage distortion filter $H_{\text{sub.IV}}(s)$ may be created on a power amplifier circuit **12** when the power amplifier circuit **12** is coupled to an RF front-end circuit **14**. Notably, in the unwanted voltage distortion filter $H_{\text{sub.IV}}(s)$, “s” is a notation of Laplace transform. The existing transmission circuit **10** includes a transceiver circuit **16**, a power management integrated circuit (PMIC) **18**, and a transmitter circuit **20**, which can include an antenna(s) (not shown) as an example.

(24) The transceiver circuit **16** is configured to generate an RF signal **22** having a time-variant input power $P_{\text{sub.IN}}(t)$ that corresponds to a time-variant voltage envelope **24** and provides the RF signal **22** to the power amplifier circuit **12**. The transceiver circuit **16** is also configured to generate a time-variant target voltage $V_{\text{sub.TGT}}$, which is associated with a time-variant target voltage envelope **26** that tracks the time-variant voltage envelope **24** of the RF signal **22**. The PMIC **18** is configured to generate a modulated voltage $V_{\text{sub.CC}}$ having a time-variant modulated voltage envelope **28** that tracks the time-variant target voltage envelope **26** of the time-variant target voltage $V_{\text{sub.TGT}}$ and provides the modulated voltage $V_{\text{sub.CC}}$ to the power amplifier circuit **12**. The power amplifier circuit **12** is configured to amplify the RF signal **22** based on the modulated voltage $V_{\text{sub.CC}}$ to a time-variant output voltage $V_{\text{sub.OUT}}$ associated with a time-variant output voltage envelope **30**. The power amplifier circuit **12** then provides the amplified RF signal **22** to the RF front-end circuit **14**. The RF front-end circuit **14** may include, for example, a filter circuit that performs further frequency filtering on the amplified RF signal **22** before providing the amplified RF signal **22** to the transmitter circuit **20** for transmission.

(25) FIG. 1B is a schematic diagram providing an exemplary illustration of an output stage **32** of the power amplifier circuit **12** in FIG. 1A. Common elements between FIGS. 1A and 1B are shown therein with common element numbers and will not be re-described herein.

(26) The output stage **32** can include at least one transistor **34**, such as a bipolar junction transistor (BJT) or a complementary metal-oxide semiconductor (CMOS) transistor. Taking the BJT as an example, the transistor **34** can include a base electrode B, a collector electrode C, and an emitter electrode E. The base electrode B is configured to receive a bias voltage $V_{\text{sub.BIAS}}$ and the

collector electrode C is configured to receive the modulated voltage V.sub.CC. The collector electrode C is also coupled to the RF front-end circuit **14** and configured to output the amplified RF signal **22** at the output voltage V.sub.OUT. In this regard, the output voltage V.sub.OUT can be a function of the modulated voltage V.sub.CC. Understandably, the power amplifier circuit **12** will operate with good efficiency and linearity when the time-variant modulated voltage envelope **28** is aligned with the time-variant output voltage envelope **30**.

(27) FIG. **1C** is a schematic diagram of an exemplary equivalent model **36** providing an exemplary illustration of the voltage distortion filter H.sub.IV(s) created by the coupling between the power amplifier circuit **12** and the RF front-end circuit **14** in the existing transmission circuit **10** of FIG. **1A**. Common elements between FIGS. **1A-1C** are shown therein with common element numbers and will not be re-described herein

(28) In the equivalent model **36**, V.sub.PA and Z.sub.PA represent the output stage **32** of the power amplifier circuit **12** and an inherent impedance of the power amplifier circuit **12**, respectively, and Z.sub.11 represents an inherent impedance associated with an input port of the RF front-end circuit **14**. Herein, V.sub.OUT represents an output voltage associated with the RF signal **22** before the power amplifier circuit **12** is coupled to the RF front-end circuit **14**, and V'.sub.OUT represents an output voltage associated with the RF signal **22** after the power amplifier circuit **12** is coupled to the RF front-end circuit **14**. Hereinafter, the output voltages V.sub.OUT and V'.sub.OUT are referred to as “non-coupled output voltage” and “coupled output voltage,” respectively, for distinction.

(29) A Laplace transform representative of the coupled output voltage V'.sub.OUT can be expressed in equation (Eq. 1) below.

$$(30) \quad V'_{OUT}(s) = \frac{V_{OUT}(s) * [1 - T_{PA}(s)] * [1 + T_I(s)]}{2 * [1 - T_{PA}(s) * T_I(s)]} = V_{OUT}(s) * H_{IV}(s) \quad (\text{Eq. 1})$$

$$H_{IV}(s) = \frac{[1 - T_{PA}(s)] * [1 + T_I(s)]}{2 * [1 - T_{PA}(s) * T_I(s)]}$$

(31) In the equation (Eq. 1) above, .sub.PA(s) represents a reflection coefficient looking back into the output stage **32** of the power amplifier circuit **12** and .sub.I(s) represents a reflection coefficient looking into the RF front-end circuit **14**. Notably, .sub.PA(s) and .sub.I(s) are complex filters containing amplitude and phase information. In this regard, the .sub.PA(s), the .sub.I(s), and, therefore, the voltage distortion filter H.sub.IV(s) are dependents of such factors as modulation bandwidth, RF frequency, and/or voltage standing wave ratio (VSWR).

(32) The equation (Eq. 1) shows that the coupled output voltage V'.sub.OUT will be altered from the non-coupled output voltage V.sub.OUT by the voltage distortion filter H.sub.IV(s) when the power amplifier circuit **12** is coupled to the RF front-end circuit **14**. Moreover, the variation of the non-coupled output voltage V.sub.OUT caused by the voltage distortion filter H.sub.IV(s) can happen across all frequencies in an entire modulation bandwidth of the RF signal **22**. As a result, the coupled output voltage V'.sub.OUT may become misaligned from the modulated voltage V.sub.CC across the modulation bandwidth of the RF signal **22**, thus causing a frequency dependent AM-AM error AM.sub.ERR across the modulation bandwidth of the existing transmission circuit **10**.

(33) With reference back to FIG. **1A**, when the power amplifier circuit **12** is coupled to the RF front-end circuit **14**, an output reflection coefficient (e.g., S.sub.22) of the power amplifier circuit **12** can interact with an input reflection coefficient (e.g., S.sub.11) of the RF front-end circuit **14** to create a group delay τ in the RF signal **22**. The group delay τ , which can be expressed in equation (Eq. 2) below, can cause a phase error $\Delta\phi$ at the power amplifier circuit **12** to thereby create an AM-PM error ϕ .sub.ERR in the RF signal **22**.

$$\tau = -\Delta\phi / \Delta t \quad (\text{Eq. 2})$$

(34) Studies have shown that the group delay τ in each of the transmission frequencies varies in accordance with the time-variant input power $P_{\text{sub.IN}}(t)$, as illustrated in FIG. 1D. FIG. 1D is a graphic diagram illustrating a distribution of the group delay τ across multiple transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ within a modulation bandwidth of the RF signal **22** in FIG. 1A. As shown in FIG. 1D, each of the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ is associated with a respective one of multiple variable group delays $\tau_{\text{sub.1}}(P_{\text{sub.IN}})-\tau_{\text{sub.M}}(P_{\text{sub.IN}})$, and each of the variable group delays $\tau_{\text{sub.1}}(P_{\text{sub.IN}})-\tau_{\text{sub.M}}(P_{\text{sub.IN}})$ varies independently in accordance with the time-variant input power $P_{\text{sub.IN}}(t)$.

(35) Given the relationship between the group delay τ and the phase error $\Delta\phi$ in equation (Eq. 2), the phase error $\Delta\phi$ associated with each of the variable group delays $\tau_{\text{sub.1}}(P_{\text{sub.IN}})-\tau_{\text{sub.M}}(P_{\text{sub.IN}})$ will also vary according to the time-variant input power $P_{\text{sub.IN}}(t)$. FIG. 1E is a graphic diagram illustrating a distribution of the variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ across the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ of the RF signal **22** in FIG. 1A. As shown in FIG. 1E, each of the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ is associated with a respective one of multiple variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$. Moreover, for any given level of the time-variant input power $P_{\text{sub.IN}}(t)$, each of the variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ associated with a respective one of the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ is nonlinear relative to any other variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ associated with any other transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$. For example, the variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})$ associated with the transmission frequency $F_{\text{sub.1}}$ is nonlinearly related to the variable phase errors $\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ associated with the transmission frequency $F_{\text{sub.M}}$ for any given value of the time-variant input power $P_{\text{sub.IN}}(t)$.

(36) Similar to the frequency dependent amplitude error $AM_{\text{sub.ERR}}$, the AM-PM error $\phi_{\text{sub.ERR}}$ resulted from the variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ are also frequency dependent. FIG. 1F is a graphic diagram illustrating a distribution of the AM-AM and AM-PM errors across the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ of the RF signal **22** in FIG. 1A.

(37) As shown in FIG. 1F, the AM-PM error $\phi_{\text{sub.ERR}}$ varies according to the modulated voltage $V_{\text{sub.CC}}$ in each of the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$. In this regard, it is necessary to correct both the AM-AM and the AM-PM errors in the existing transmission circuit **10** to ensure proper alignment between the time-variant voltage envelope **24**, the time-variant modulated voltage envelope **28**, and the time-variant output voltage envelope **30**.

(38) In this regard, FIG. 2 is a schematic diagram of an exemplary transmission circuit **38** that can be configured according to embodiments of the present disclosure to correct the AM-AM $AM_{\text{sub.ERR}}$ and the AM-PM error $\phi_{\text{sub.ERR}}$ in the existing transmission circuit **10** of FIG. 1A across a modulation bandwidth that includes multiple transmission frequencies (a.k.a. target frequencies) $F_{\text{sub.1}}-F_{\text{sub.M}}$. Herein, the transmission circuit **38** includes a transceiver circuit **40**, a PMIC **42**, and a power amplifier circuit **44**.

(39) The transceiver circuit **40** is configured to generate an RF signal **46** in a time-variant input power $P_{\text{sub.IN}}(t)$ and for transmission in a selected transmission frequency (a.k.a. selected target frequency) $F_{\text{sub.TGT}}$ among the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$. The power amplifier circuit **44** is configured to amplify the RF signal **46** from the time-variant input power $P_{\text{sub.IN}}(t)$ to a time-variant output power $P_{\text{sub.OUT}}(t)$ based on a modulated voltage $V_{\text{sub.CC}}$ to thereby generate an amplified RF signal **46 AMP**. The PMIC **42** is configured to generate the modulated voltage $V_{\text{sub.CC}}$, which can be an envelope tracking (ET) modulated voltage or an average power tracking (APT) modulated voltage, based on a modulated target voltage $V_{\text{sub.TGT}}$. The transceiver circuit **40** is also configured to generate the modulated target voltage $V_{\text{sub.TGT}}$ that tracks the time-variant input power $P_{\text{sub.IN}}(t)$ and provides the modulated target voltage $V_{\text{sub.TGT}}$ to the PMIC **42**. Ideally, since the modulated target voltage $V_{\text{sub.TGT}}$ tracks the time-variant input power $P_{\text{sub.IN}}(t)$ and the modulated voltage $V_{\text{sub.CC}}$ tracks the modulated target voltage

V.sub.TGT, the modulated voltage V.sub.CC should be in good alignment with the time-variant input power P.sub.IN(t). In this regard, if the power amplifier circuit **44** can operate with a linear gain, the time-variant output power P.sub.OUT(t) should also be linearly related to the time-variant input power P.sub.IN(t) by the linear gain of the power amplifier circuit **44**.

(40) Like the power amplifier circuit **12** in FIG. 1A, the power amplifier circuit **44** can also include the output stage **32** (as shown in FIG. 1B) and be coupled to an RF front-end circuit **48** (e.g., a filter, multiplexer, and/or switches).

(41) As such, the transmission circuit **38** may also suffer the frequency dependent AM-AM error AM.sub.ERR and the frequency dependent AM-PM error ϕ .sub.ERR as in the existing transmission circuit **10** of FIG. 1A. In other words, in the transmission circuit **38**, the output reflection coefficient (e.g., S.sub.22) of the power amplifier circuit **44** can interact with the input reflection coefficient (e.g., S.sub.11) of the RF front-end circuit **48** to create the voltage distortion filter H.sub.IV(s) and, consequently, cause the AM-AM error AM.sub.ERR and the AM-PM error ϕ .sub.ERR.

(42) As described in detail below, the transmission circuit **38** can be configured according to various embodiments of the present disclosure to effectively correct the AM-AM error AM.sub.ERR and the AM-PM error ϕ .sub.ERR. More specifically, the transmission circuit **38** can concurrently correct the AM-AM and AM-PM errors via multiple complex filters. As a result, the transmission circuit **38** can reduce undesired instantaneous excessive compression and/or spectrum regrowth to thereby improve efficiency and linearity of the power amplifier circuit **44** across an entire modulation bandwidth of the transmission circuit **38**.

(43) In an embodiment, the power amplifier circuit **44** includes a phase shifter **50** and a power amplifier **52**. The phase shifter **50** is configured to apply a phase shift $\Delta\phi$ to the RF signal **46** to correct the AM-PM error ϕ .sub.ERR. More specifically, the phase shifter **50** is configured to receive a modulated phase correction voltage V.sub. ϕ from the PMIC **42** and determine the phase shift $\Delta\phi$ based on the modulated phase correction voltage V.sub. ϕ . In a non-limiting example, the phase shifter **50** may include internal storage (not shown), such as registers for example, to store a correlation between various levels of the modulated phase correction voltage V.sub. ϕ and various degrees of phase shift $\Delta\phi$. For example, the phase shifter **50** can store a correlation between the modulated phase correction voltage V.sub. ϕ of 0 V, 1 V, and 2 V and the phase shift $\Delta\phi$ of 0°, 1°, and 2°, respectively. Accordingly, the phase shifter **50** can determine the correct phase shift $\Delta\phi$ based on the modulated phase correction voltage V.sub. ϕ and apply the determined phase shift $\Delta\phi$ to the RF signal **46** to generate a phase-shifted RF signal **46 ϕ** . Understandably, since the phase shifter **50** only applies the phase shift $\Delta\phi$ to the RF signal **46**, the phase-shifted RF signal **46 ϕ** will have the same time-variant input power P.sub.IN(t) and in the same selected target frequency F.sub.TGT.

(44) Subsequently, the power amplifier **52** can amplify the phase-shifted RF signal **46 ϕ** based on a modulate voltage, such as an envelope tracking (ET) voltage or an average power tracking (APT) voltage, to generate the amplified RF signal **46AMP**. Understandably, since the power amplifier **52** only changes the time-variant input power P.sub.IN(t) in the phase-shifted RF signal **46 ϕ** to the time-variant output power P.sub.OUT(t) in the amplified RF signal **46AMP**, the amplified RF signal **46AMP** will maintain the same phase as in the phase-shifted RF signal **46 ϕ** . Thus, by generating the modulated phase correction voltage V.sub. ϕ at a proper voltage level, it is possible to correct the AM-PM error ϕ .sub.ERR in the amplified RF signal **46AMP**.

(45) The PMIC **42** includes a voltage modulation circuit **54** and a phase correction circuit **56**. According to an embodiment of the present disclosure, the voltage modulation circuit **54** is configured to generate the modulated voltage V.sub.CC based on the modulated target voltage V.sub.TGT and provide the modulated voltage V.sub.CC to the power amplifier **52**. The phase correction circuit **56** is configured to generate the phase correction voltage V.sub. ϕ based on the modulated target voltage V.sub.TGT and provide the phase correction voltage V.sub. ϕ to the phase

shifter 50.

(46) In an embodiment, the transceiver circuit 40 is configured to generate the modulated voltage $V_{sub.TGT}$ and provide the modulated target voltage $V_{sub.TGT}$ to the voltage modulation circuit 54 and the phase correction circuit 56. In this regard, FIG. 3 is a schematic diagram illustrating the transceiver circuit 40 configured according to an embodiment of the present disclosure. Common elements between FIGS. 2 and 3 are shown therein with common element numbers and will not be re-described herein.

(47) Herein, the transceiver circuit 40 includes a digital processing circuit 58, a delay equalizer circuit 60, an amplitude correction circuit 62, a target voltage circuit 64, and a signal conversion circuit 66. The digital processing circuit 58, which can be a digital baseband circuit as an example, is configured to generate an input vector  custom character corresponding to a time-variant amplitude $AM(t)$. The input vector  custom character may be modulated in a baseband frequency that can be converted to any of the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$ within the modulation bandwidth of the transmission circuit 38.

(48) According to the previous discussion in FIG. 1D, the input vector  custom character can be associated with multiple variable group delays $\tau_{sub.1}(P_{sub.IN})$ - $\tau_{sub.M}(P_{sub.IN})$, each corresponding to a respective one of the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$. In other words, the variable group delays $\tau_{sub.1}(P_{sub.IN})$ - $\tau_{sub.M}(P_{sub.IN})$ can be different from one another between the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$.

(49) The delay equalizer circuit 60 is configured to apply a delay equalization filter $H_{sub.\tau}(s)$ to the input vector  custom character to convert each of the variable group delays $\tau_{sub.1}(P_{sub.IN})$ - $\tau_{sub.M}(P_{sub.IN})$ into a respective one of multiple constant group delays $\tau_{sub.1}$ - $\tau_{sub.M}$, as illustrated in FIG. 4A. FIG. 4A is a graphic diagram illustrating a distribution of the constant group delays $\tau_{sub.1}$ - $\tau_{sub.M}$ across the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$ as a result of applying the delay equalization filter $H_{sub.\tau}(s)$ to the input vector  custom character. As shown in FIG. 4A, for each of the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$, the respective one of the group delays $\tau_{sub.1}$ - $\tau_{sub.M}$ stays constant relative to the time-variant input power $P_{sub.IN}(t)$.

(50) Given the constant group delays $\tau_{sub.1}$ - $\tau_{sub.M}$ and the τ - $\Delta\phi$ relationship established in equation (Eq. 2), the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ - $\Delta\phi_{sub.M}(P_{sub.IN})$ will in turn exhibit a linear relationship across the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$, as illustrated in FIG. 4B.

(51) FIG. 4B is a graphic diagram illustrating a distribution of the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ - $\Delta\phi_{sub.M}(P_{sub.IN})$ across the transmission frequencies $F_{sub.1}$ - $F_{sub.M}$ as a result of applying the delay equalization filter $H_{sub.\tau}(s)$ to the input vector  custom character. Given the linear relationship between the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ - $\Delta\phi_{sub.M}(P_{sub.IN})$, each of the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ - $\Delta\phi_{sub.M}(P_{sub.IN})$ can be moved up or down based on an appropriate scaling factor $F_{sub.SCALE}$ to superimpose on another one of the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ - $\Delta\phi_{sub.M}(P_{sub.IN})$. For example, the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ associated with the transmission frequency $F_{sub.1}$ can be moved downward to superimpose on the variable phase errors $\Delta\phi_{sub.M}(P_{sub.IN})$ associated with the transmission frequency $F_{sub.M}$. Likewise, the variable phase errors $\Delta\phi_{sub.M}(P_{sub.IN})$ associated with the transmission frequency $F_{sub.M}$ can be moved upward to superimpose on the variable phase errors $\Delta\phi_{sub.1}(P_{sub.IN})$ associated with the transmission frequency $F_{sub.1}$.

(52) With reference back to FIG. 3, by applying the delay equalization filter $H_{sub.\tau}(s)$ to the input vector  custom character, the delay equalizer circuit 60 generates a delay-equalized vector  custom character. Specifically, the delay-equalized vector  custom character can be associated with a respective one of the constant group delays $\tau_{sub.1}$ - $\tau_{sub.M}$ corresponding to the selected target frequency $F_{sub.TGT}$. The signal conversion circuit 66, which may include, for example, a digital-to-analog converter (DAC) and a frequency converter, can then convert the delay-equalized vector  custom character in the RF signal 46 in the selected target frequency $F_{sub.TGT}$. In an

embodiment, the digital processing circuit **58** may be configured to generate an indication **68** that indicates the selected target frequency $F_{\text{sub.TGT}}$ and provide the indication **68** to the phase correction circuit **56**.

(53) To correct the AM-AM error $AM_{\text{sub.ERR}}$, the amplitude correction circuit **62** is configured to equalize the delay-equalized vector  to thereby generate a delay-gain-equalized vector  having a constant gain in the selected target frequency $F_{\text{sub.TGT}}$. The target voltage circuit **64**, in turn, generates the modulated target voltage $V_{\text{sub.TGT}}$ from the delay-gain-equalized vector  and provides the modulated target voltage $V_{\text{sub.TGT}}$ to the PMIC **42**. For an in-depth description of the amplitude correction circuit **62**, please refer to U.S. patent application Ser. No. 17/700,826, entitled “ENVELOPE TRACKING VOLTAGE CORRECTION IN A TRANSMISSION CIRCUIT.”

(54) With reference back to FIG. 2, to correct the AM-PM error $\phi_{\text{sub.ERR}}$, the phase correction circuit **56** is configured to explore the linearity and scalability characteristics of variable phase errors $\Delta\phi_{\text{sub.1}}(P_{\text{sub.IN}})-\Delta\phi_{\text{sub.M}}(P_{\text{sub.IN}})$ across the transmission frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$, as illustrated in FIG. 4B, to thereby determine the modulated phase correction voltage $V_{\text{sub.}\phi}$. In an embodiment, the phase correction circuit **56** includes a phase correction voltage lookup table (LUT) circuit **70** and a scaling circuit **72**. The phase correction voltage LUT circuit **70** may include a phase correction voltage LUT (not shown) that correlates the modulated target voltage $V_{\text{sub.TGT}}$ with various levels of a reference phase correction voltage $V_{\text{sub.}\phi\text{-REF}}$ predefined for a reference frequency $F_{\text{sub.REF}}$. In a non-limiting example, the reference frequency $F_{\text{sub.REF}}$ can be any of the target frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$, a center frequency of the modulation bandwidth of the transmission circuit **38**, or even an arbitrary frequency. Moreover, the reference frequency $F_{\text{sub.REF}}$ may be identical to or different from the selected target frequency $F_{\text{sub.TGT}}$. In this regard, the phase correction voltage LUT circuit **70** can look up the phase correction LUT to select the reference phase correction voltage $V_{\text{sub.}\phi\text{-REF}}$ based on the modulated target voltage $V_{\text{sub.TGT}}$.

(55) FIG. 5 is a schematic diagram illustrating the scaling circuit **72** in the phase correction circuit **56** in FIG. 2. Common elements between FIGS. 2 and 5 are shown therein with common element numbers and will not be re-described herein.

(56) In an embodiment, the scaling circuit **72** includes a scaling LUT circuit **74** and a multiplier **76**. The scaling LUT circuit **74** is configured to determine the scaling factor $F_{\text{sub.SCALE}}$ corresponding to the selected target frequency $F_{\text{sub.TGT}}$ based on, for example, a scaling LUT (not shown) stored in the scaling LUT circuit **74**. The multiplier **76** is configured to multiply the reference phase correction voltage $V_{\text{sub.}\phi\text{-REF}}$ by the scaling factor $F_{\text{sub.SCALE}}$ to thereby generate the phase correction voltage $V_{\text{sub.}\phi}$ ($V_{\text{sub.}\phi}=V_{\text{sub.}\phi\text{-REF}}\times F_{\text{sub.SCALE}}$).

(57) The transmission circuit **38** of FIG. 2 can be configured to correct the AM-AM error $AM_{\text{sub.ERR}}$ and the AM-PM error $\phi_{\text{sub.ERR}}$ based on a process. In this regard, FIG. 6 is a flowchart of an exemplary process **200** that can be employed by the transmission circuit of FIG. 2 to correct the AM-AM error $AM_{\text{sub.ERR}}$ and the AM-PM error $\phi_{\text{sub.ERR}}$.

(58) Herein, the phase correction circuit **56** receives the modulated target voltage $V_{\text{sub.TGT}}$ and the indication **68** that indicates the selected target frequency $F_{\text{sub.TGT}}$ among the target frequencies $F_{\text{sub.1}}-F_{\text{sub.M}}$ (step **202**). Accordingly, the phase correction circuit **56** determines the reference phase correction voltage $V_{\text{sub.}\phi\text{-REF}}$ corresponding to a reference frequency $F_{\text{sub.REF}}$ based on the modulated target voltage $V_{\text{sub.TGT}}$ (step **204**). Next, the phase correction circuit **56** determines the scaling factor $F_{\text{sub.SCALE}}$ corresponding to the selected target frequency $F_{\text{sub.TGT}}$ (step **206**). Subsequently, the phase correction circuit **56** generates the phase correction voltage $V_{\text{sub.}\phi}$ based on the determined reference phase correction voltage $V_{\text{sub.}\phi\text{-REF}}$ and the determined scaling factor $F_{\text{sub.SCALE}}$ (step **208**). The phase shifter **50** determines the phase shift $\Delta\phi$ based on the phase correction voltage $V_{\text{sub.}\phi}$ (step **210**). Accordingly, the phase shifter applies the phase shift $\Delta\phi$ to the RF signal **46**, which is modulated for transmission in the selected target

frequency F.sub.TGT, to thereby generate a phase-shifted RF signal 46ϕ (step 212).
(59) Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A transmission circuit comprising: a power management integrated circuit (PMIC) comprising a phase correction circuit configured to: receive a modulated target voltage tracking a time-variant input power of a radio frequency (RF) signal and an indication of a selected target frequency among a plurality of target frequencies for transmission of the RF signal; determine a reference phase correction voltage predefined for a reference frequency based on the modulated target voltage; determine a scaling factor corresponding to the selected target frequency; and scale the determined reference phase correction voltage by the determined scaling factor to thereby generate a phase correction voltage; and a power amplifier circuit comprising a phase shifter configured to: determine a phase shift corresponding to the phase correction voltage; and apply the phase shift to the RF signal to thereby generate a phase-shifted RF signal.
2. The transmission circuit of claim 1, wherein: the power amplifier circuit further comprises a power amplifier configured to amplify the phase-shifted RF signal based on a modulated voltage for transmission in the respective one of the plurality of target frequencies; and the PMIC further comprises a voltage modulation circuit configured to generate the modulated voltage based on the modulated target voltage.
3. The transmission circuit of claim 1, wherein the phase correction circuit comprises: a phase correction voltage lookup table (LUT) circuit configured to determine the reference phase correction voltage predefined for the reference frequency based on the modulated target voltage; and a scaling circuit configured to: determine the scaling factor corresponding to the selected target frequency; and multiply the reference phase correction voltage by the scaling factor to thereby generate the phase correction voltage.
4. The transmission circuit of claim 3, wherein the scaling circuit comprises: a scaling LUT circuit configured to determine the scaling factor corresponding to the selected target frequency; and a multiplier configured to multiply the reference phase correction voltage by the scaling factor to generate the phase correction voltage.
5. The transmission circuit of claim 1, wherein the reference frequency is identical to the selected target frequency among the plurality of target frequencies.
6. The transmission circuit of claim 1, wherein the reference frequency is any one of the plurality of target frequencies.
7. The transmission circuit of claim 1, further comprising a transceiver circuit configured to: generate the RF signal based on an input vector and having the time-variant input power corresponding to a time-variant amplitude of the input vector; and generate the modulated target voltage based on the time-variant amplitude of the input vector.
8. The transmission circuit of claim 7, wherein the transceiver circuit comprises: a digital processing circuit configured to generate the input vector associated with a plurality of time-variant group delays each corresponding to a respective one of the plurality of target frequencies; and a delay equalizer circuit configured to: equalize the input vector based on a delay equalization filter to thereby convert the plurality of time-variant group delays into a plurality of constant group delays each corresponding to the respective one of the plurality of target frequencies; and generate a delay-equalized vector associated with a respective one of the plurality of constant group delays corresponding to the selected target frequency among the plurality of target frequencies; and a signal conversion circuit configured to generate the RF signal for transmission in the selected target frequency based on the delay-equalized vector.

9. The transmission circuit of claim 8, wherein the digital processing circuit is further configured to generate the indication of the selected target frequency among the plurality of target frequencies.
10. The transmission circuit of claim 8, wherein the transceiver circuit further comprises: an amplitude correction circuit configured to equalize the delay-equalized vector to thereby generate a delay-gain-equalized vector having a constant gain in the selected target frequency; and a target voltage circuit configured to generate the modulated target voltage based on the delay-gain-equalized vector.
11. A method for correcting phase and amplitude errors in a transmission circuit comprising: receiving a modulated target voltage tracking a time-variant input power of a radio frequency (RF) signal and an indication of a selected target frequency among a plurality of target frequencies for transmission of the RF signal; determining a reference phase correction voltage predefined for a reference frequency based on the modulated target voltage; determining a scaling factor corresponding to the selected target frequency; scaling the determined reference phase correction voltage by the determined scaling factor to thereby generate a phase correction voltage; determining a phase shift corresponding to the phase correction voltage; and applying the phase shift to the RF signal to thereby generate a phase-shifted RF signal.
12. The method of claim 11, further comprising: amplifying the phase-shifted RF signal based on a modulated voltage for transmission in the respective one of the plurality of target frequencies; and generating the modulated voltage based on the modulated target voltage.
13. The method of claim 11, further comprising: determining, from a phase correction voltage lookup table (LUT), the reference phase correction voltage predefined for the reference frequency based on the modulated target voltage; and multiplying the reference phase correction voltage by the scaling factor to thereby generate the phase correction voltage.
14. The method of claim 11, further comprising selecting the reference frequency to be: identical to the selected target frequency among the plurality of target frequencies.
15. The method of claim 11, further comprising selecting the reference frequency to be any one of the plurality of target frequencies.
16. The method of claim 11, further comprising: generating the RF signal based on an input vector and having the time-variant input power corresponding to a time-variant amplitude of the input vector; and generating the modulated target voltage based on the time-variant amplitude of the input vector.
17. The method of claim 16, further comprising: generating the input vector associated with a plurality of time-variant group delays each corresponding to a respective one of the plurality of target frequencies; equalizing the input vector based on a delay equalization filter to thereby convert the plurality of time-variant group delays into a plurality of constant group delays each corresponding to the respective one of the plurality of target frequencies; generating a delay-equalized vector associated with a respective one of the plurality of constant group delays corresponding to the selected target frequency among the plurality of target frequencies; and generating the RF signal for transmission in the selected target frequency based on the delay-equalized vector.
18. The method of claim 17, further comprising: equalizing the delay-equalized vector to thereby generate a delay-gain-equalized vector having a constant gain in the selected target frequency; and generating the modulated target voltage based on the delay-gain-equalized vector.
19. The transmission circuit of claim 1, wherein the reference frequency is a center frequency of a modulation bandwidth of the transmission circuit.
20. A wireless device comprising a transmission circuit, the transmission circuit comprises: a power management integrated circuit (PMIC) comprising a phase correction circuit configured to: receive a modulated target voltage tracking a time-variant input power of a radio frequency (RF) signal and an indication of a selected target frequency among a plurality of target frequencies for transmission of the RF signal; determine a reference phase correction voltage predefined for a reference

frequency based on the modulated target voltage; determine a scaling factor corresponding to the selected target frequency; and scale the determined reference phase correction voltage by the determined scaling factor to thereby generate a phase correction voltage; and a power amplifier circuit comprising a phase shifter configured to: determine a phase shift corresponding to the phase correction voltage; apply the phase shift to the RF signal to thereby generate a phase-shifted RF signal; and amplify the phase-shifted RF signal based on a modulated voltage for transmission in the selected target frequency.
